

| space                         | probe | per-ev. | $f^*(2\sigma)$ , median over draws |       |         | SparKer range |
|-------------------------------|-------|---------|------------------------------------|-------|---------|---------------|
|                               |       |         | Maha.                              | MMD   | SparKer |               |
| SupCon                        | 0.930 | 0.01    | 0.057                              | 0.020 | 0.018   | 0.015–0.021   |
| → res (residual)              | 0.923 | 0.23    | 0.023                              | 0.033 | 0.022   | 0.012–0.039   |
| → res (concat)                | 0.957 | 0.12    | 0.024                              | 0.018 | 0.019   | 0.016–0.021   |
| → res-nplm (residual)         | 0.926 | 0.22    | 0.023                              | 0.029 | 0.017   | 0.015–0.019   |
| → res-nplm (concat)           | 0.946 | 0.10    | 0.025                              | 0.020 | 0.019   | 0.016–0.021   |
| SupCon+SIGReg ( $\lambda=5$ ) | 0.912 | 0.02    | $> 0.1^4$                          | 0.031 | 0.037   | 0.032–0.042   |
| → res (residual)              | 0.902 | 0.01    | $> 0.1^3$                          | 0.044 | 0.069   | 0.044–0.091   |
| → res (concat)                | 0.949 | 0.01    | $> 0.1^3$                          | 0.035 | 0.042   | 0.027–0.060   |
| → res-nplm (residual)         | 0.921 | 0.16    | 0.044                              | 0.038 | 0.022   | 0.017–0.025   |
| → res-nplm (concat)           | 0.943 | 0.06    | $> 0.1^2$                          | 0.033 | 0.030   | 0.020–0.036   |
| NPLM-dist. (sup.)             | 0.788 | 0.17    | 0.038                              | 0.039 | 0.027   | 0.024–0.040   |
| → res (residual)              | 0.884 | 0.03    | $> 0.1^2$                          | 0.046 | 0.046   | 0.044–0.077   |
| → res (concat)                | 0.908 | 0.03    | $> 0.1^2$                          | 0.042 | 0.043   | 0.033–0.059   |
| → res-nplm (residual)         | 0.841 | 0.01    | $> 0.1^2$                          | 0.044 | 0.039   | 0.024–0.049   |
| → res-nplm (concat)           | 0.879 | 0.09    | 0.047 <sup>1</sup>                 | 0.045 | 0.031   | 0.018–0.043   |
| NPLM-dist.+classwise SIGReg   | 0.938 | 0.21    | 0.032                              | 0.033 | 0.027   | 0.024–0.029   |
| → res (residual)              | 0.917 | 0.03    | $> 0.1^2$                          | 0.037 | 0.048   | 0.039–0.084   |
| → res (concat)                | 0.958 | 0.03    | 0.034 <sup>1</sup>                 | 0.035 | 0.047   | 0.037–0.078   |
| → res-nplm (residual)         | 0.948 | 0.07    | 0.034                              | 0.036 | 0.043   | 0.028–0.079   |
| → res-nplm (concat)           | 0.966 | 0.17    | 0.023                              | 0.035 | 0.037   | 0.018–0.046   |

**Table 1: Pre-discovery: minimum injected fraction for a  $2\sigma$  dataset-level detection**, with and without the residual construction.  $f^*$  is the smallest fraction of the held-out class injected into a 5000-point corpus at which the median expected significance of the test reaches  $2\sigma$ ; lower is better. The probe (oracle) and per-event power are shown for reference; the last column is SparKer’s range across draws. *Coverage*: CIFAR-10, leakage-free lineage, single holdout; draws 4, 7, 8, 9; 20/24 spaces on every draw; 200 null / 50 signal toys per fraction,  $N_D=5000$ ; median over draws, superscript = draws where the test never reached  $2\sigma$  by  $f=0.1$ .

## Sigma-currency tables (exps 146/148), CIFAR-10 leakage-free lineage — generated September 6, 2026

$f^*(2\sigma)$  = smallest injected fraction at which the median expected significance of a dataset-level test reaches  $2\sigma$  (200 null / 50 signal toys,  $N_D=5000$ ); lower is better. Post tables: bold = best test per arm; † = the derived cut declined at some fraction below the crossing (the space is the frozen one there); purity ‘–’ = declined at  $f=0.03$ ; ‘(n/6)’ = arm still running.

| arm                           | draw | frozen SparKer | dist/95th                      | dist/derived                   | np/derived                   |
|-------------------------------|------|----------------|--------------------------------|--------------------------------|------------------------------|
| SupCon                        | h4   | 0.015          | 0.027 (MMD)                    | 0.014 (MMD) <sup>†</sup>       | 0.014 (MMD) <sup>†</sup>     |
|                               | h7   | 0.021          | 0.035 (SparKer (anch.))        | 0.033 (SparKer (anch.))        | <b>0.018 (SparKer)</b>       |
|                               | h8   | 0.018          | <b>0.016 (Eucl. (anch.))</b>   | <b>0.017 (MMD)</b>             | <b>0.011 (Eucl. (anch.))</b> |
|                               | h9   | 0.018          | <b>0.008 (Eucl.)</b>           | <b>0.011 (Eucl. (anch.))</b>   | <b>0.009 (SparKer)</b>       |
| SupCon+SIGReg ( $\lambda=5$ ) | h4   | 0.042          | <b>0.021 (Eucl. (anch.))</b>   | <b>0.028 (MMD)</b>             | <b>0.028 (Eucl.)</b>         |
|                               | h7   | 0.034          | <b>0.021 (Eucl. (anch.))</b>   | 0.036 (SparKer) <sup>†</sup>   | 0.036 (SparKer) <sup>†</sup> |
|                               | h8   | 0.040          | <b>0.015 (MMD)</b>             | <b>0.021 (MMD)</b>             | <b>0.020 (Eucl. (anch.))</b> |
|                               | h9   | 0.032          | <b>0.016 (Eucl. (anch.))</b>   | 0.026 (MMD) <sup>†</sup>       | 0.026 (MMD) <sup>†</sup>     |
| NPLM-dist. (sup.)             | h4   | 0.040          | <b>0.024 (Eucl. (anch.))</b>   | <b>0.037 (Eucl. (anch.))</b>   | <b>0.033 (Eucl. (anch.))</b> |
|                               | h7   | 0.029          | <b>0.022 (Eucl. (anch.))</b>   | 0.039 (Eucl. (anch.))          | <b>0.021 (Eucl. (anch.))</b> |
|                               | h8   | 0.024          | <b>0.016 (SparKer (anch.))</b> | <b>0.015 (Eucl. (anch.))</b>   | <b>0.016 (Eucl. (anch.))</b> |
|                               | h9   | 0.024          | <b>0.009 (SparKer (anch.))</b> | <b>0.020 (Eucl. (anch.))</b>   | <b>0.020 (Eucl. (anch.))</b> |
| NPLM-dist.+classwise SIGReg   | h4   | 0.025          | <b>0.019 (Eucl. (anch.))</b>   | <b>0.020 (Eucl. (anch.))</b>   | <b>0.020 (Eucl. (anch.))</b> |
|                               | h7   | 0.029          | <b>0.018 (Eucl. (anch.))</b>   | <b>0.027 (SparKer (anch.))</b> | <b>0.011 (Eucl. (anch.))</b> |
|                               | h8   | 0.024          | <b>0.015 (Eucl. (anch.))</b>   | <b>0.012 (Eucl. (anch.))</b>   | <b>0.009 (Eucl. (anch.))</b> |
|                               | h9   | 0.028          | <b>0.016 (Maha.)</b>           | <b>0.020 (Eucl. (anch.))</b>   | <b>0.020 (Eucl. (anch.))</b> |

**Table 2: Does the discovery loop beat the frozen space?** Frozen-space SparKer  $f^*(2\sigma)$  against the best post-discovery test (named in parentheses) under three loop variants: the campaign’s distance pool with the 95th-percentile cut, the distance pool with the derived cut, and the density-ratio pool with the derived cut (the paper’s construction). Bold = the loop wins on that draw. *Coverage:* CIFAR-10, four parents x draws 4, 7, 8, 9; 48 complete (arm, draw, variant) cells, the loop beats the frozen space in 38; <sup>†</sup> = the derived cut had declined at the crossing fraction, so that number is the frozen space’s and is not counted as a win; ‘-’ = variant not run or still running.

| dataset / backbone          | best frozen (space / test, median $f^*$ ) |       | best discovery (arm / test)   |       | gain           |
|-----------------------------|-------------------------------------------|-------|-------------------------------|-------|----------------|
| CIFAR-10 (scratch, 4 draws) | SupCon→res-nplm (resid.) / SparKer        | 0.017 | SupCon / Eucl. (anch.)        | 0.013 | 1.3×           |
| Galaxy10 / lejepa           | SupCon / MMD                              | 0.023 | SupCon+SIGReg / Eucl. (anch.) | 0.016 | 1.4×           |
| Galaxy10 / visreg           | SupCon→res (concat) / MMD                 | 0.026 | SupCon+SIGReg / Eucl. (anch.) | 0.017 | 1.5×           |
| Galaxy10 / dino             | SupCon+SIGReg / MMD                       | 0.029 | SupCon / Eucl. (anch.)        | 0.019 | 1.5×           |
| transductive ref. (lejepa)  | gcd-ft / Maha.                            | 0.013 | –                             | –     | sees the class |
| transductive ref. (visreg)  | gcd-ft / Maha.                            | 0.015 | –                             | –     | sees the class |
| transductive ref. (dino)    | gcd-ft / MMD                              | 0.028 | –                             | –     | sees the class |

**Table 3: The chosen algorithm, per dataset: best frozen space vs. best discovery pipeline in the same currency.** Discovery wins everywhere, always through an anchor-aware test; the transductive GCD rows are the ceiling given the whole novel class unlabelled. *Coverage:* Medians over draws (censored cells imputed at 0.15); CIFAR-10 discovery = np pool + derived cut ( $n_{\min}=10$ ), Galaxy10 discovery = np pool + derived cut ( $n_{\min}=5$ ), both skip-on-refuse; frozen candidates exclude the transductive GCD arms (shown separately).

| arm                           | frozen SparKer | purity | Eucl. | Eucl. (anch.) | Maha.        | MMD          | SparKer | SparKer (anch.) |
|-------------------------------|----------------|--------|-------|---------------|--------------|--------------|---------|-----------------|
| SupCon                        | 0.015          | 0.00   | 0.063 | > 0.1         | 0.072        | <b>0.027</b> | 0.036   | 0.029           |
| → res (residual)              | 0.039          | 0.04   | 0.074 | 0.026         | 0.039        | 0.043        | 0.047   | <b>0.026</b>    |
| → res-nplm (residual)         | 0.016          | 0.08   | 0.082 | <b>0.029</b>  | 0.065        | 0.045        | 0.037   | 0.031           |
| SupCon+SIGReg ( $\lambda=5$ ) | 0.042          | 0.01   | 0.033 | <b>0.021</b>  | > 0.1        | 0.027        | 0.031   | 0.039           |
| → res (residual)              | 0.091          | 0.01   | 0.068 | 0.043         | <b>0.035</b> | 0.036        | 0.044   | 0.045           |
| → res-nplm (residual)         | 0.023          | 0.10   | 0.027 | <b>0.017</b>  | 0.077        | 0.028        | 0.029   | 0.037           |
| NPLM-dist. (sup.)             | 0.040          | 0.10   | > 0.1 | <b>0.024</b>  | 0.087        | 0.046        | 0.046   | 0.043           |
| → res (residual)              | 0.077          | 0.01   | > 0.1 | > 0.1         | 0.081        | <b>0.035</b> | 0.043   | 0.041           |
| → res-nplm (residual)         | 0.049          | 0.01   | > 0.1 | 0.047         | 0.087        | 0.062        | 0.083   | <b>0.044</b>    |
| NPLM-dist.+classwise SIGReg   | 0.025          | 0.20   | 0.034 | <b>0.019</b>  | 0.033        | 0.044        | 0.028   | 0.024           |
| → res (residual)              | 0.084          | 0.01   | 0.027 | <b>0.027</b>  | 0.033        | 0.033        | 0.029   | 0.027           |
| → res-nplm (residual)         | 0.079          | 0.03   | 0.043 | <b>0.022</b>  | 0.037        | 0.036        | 0.047   | 0.041           |

**Table 4: Post-discovery  $f^*(2\sigma)$  on the distance pool, 95th-pct. cut.** The frozen-space SparKer  $f^*$  is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of  $d_{\text{seen}} - d_{\text{disc}}$  and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 4, distance pool, 95th-pct. cut; 12 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys,  $N_D=5000$ ; purity = round-1 pool purity at  $f=0.03$ ; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at  $f=0.03$ ; an ‘(n/m)’ row is an arm still running.

| arm                           | frozen SparKer | purity | Eucl.  | Eucl. (anch.) | Maha. | MMD            | SparKer        | SparKer (anch.) |
|-------------------------------|----------------|--------|--------|---------------|-------|----------------|----------------|-----------------|
| SupCon                        | 0.015          | –      | > 0.1  | –             | > 0.1 | <b>0.014</b> † | 0.017†         | –               |
| → res (residual)              | 0.039          | 0.03   | 0.042  | 0.039         | 0.068 | 0.039          | 0.040          | <b>0.032</b>    |
| → res-nplm (residual)         | 0.016          | 0.01   | 0.047  | 0.039         | 0.049 | 0.034          | 0.027          | <b>0.024</b>    |
| SupCon+SIGReg ( $\lambda=5$ ) | 0.042          | 0.00   | 0.038  | 0.062         | > 0.1 | <b>0.028</b>   | 0.047          | 0.044           |
| → res (residual)              | 0.091          | 0.00   | > 0.1  | > 0.1         | 0.078 | <b>0.038</b>   | 0.049          | 0.087           |
| → res-nplm (residual)         | 0.023          | 0.07   | 0.072  | <b>0.024</b>  | > 0.1 | 0.039          | 0.040          | 0.026           |
| NPLM-dist. (sup.)             | 0.040          | –      | 0.077  | <b>0.037</b>  | 0.072 | 0.045          | 0.043          | 0.043           |
| → res (residual)              | 0.077          | 0.02   | > 0.1  | > 0.1         | > 0.1 | <b>0.038</b>   | 0.052          | 0.060           |
| → res-nplm (residual)         | 0.049          | –      | 0.089  | 0.065         | 0.089 | 0.080          | <b>0.049</b> † | 0.074           |
| NPLM-dist.+classwise SIGReg   | 0.025          | –      | 0.028† | <b>0.020</b>  | 0.074 | 0.029†         | 0.028†         | 0.021           |
| → res (residual)              | 0.084          | –      | > 0.1  | 0.050         | 0.083 | <b>0.041</b>   | 0.047          | 0.050           |
| → res-nplm (residual)         | 0.079          | –      | 0.049  | 0.050         | 0.046 | <b>0.043</b>   | 0.067          | 0.050           |

**Table 5: Post-discovery  $f^*(2\sigma)$  on the distance pool, derived cut.** The frozen-space SparKer  $f^*$  is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of  $d_{\text{seen}} - d_{\text{disc}}$  and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 4, distance pool, derived cut; 12 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys,  $N_D=5000$ ; purity = round-1 pool purity at  $f=0.03$ ; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at  $f=0.03$ ; an ‘(n/m)’ row is an arm still running.

| arm                           | frozen SparKer | purity | Eucl.              | Eucl. (anch.) | Maha. | MMD                       | SparKer            | SparKer (anch.) |
|-------------------------------|----------------|--------|--------------------|---------------|-------|---------------------------|--------------------|-----------------|
| SupCon                        | 0.015          | –      | > 0.1              | –             | > 0.1 | <b>0.014</b> <sup>†</sup> | 0.017 <sup>†</sup> | –               |
| → res (residual)              | 0.039          | 0.21   | 0.033              | <b>0.015</b>  | 0.024 | 0.041                     | 0.033              | 0.029           |
| → res-nplm (residual)         | 0.016          | 0.14   | 0.035              | <b>0.020</b>  | 0.039 | 0.041                     | 0.037              | 0.034           |
| SupCon+SIGReg ( $\lambda=5$ ) | 0.042          | 0.13   | <b>0.028</b>       | 0.030         | 0.083 | 0.034                     | 0.042              | 0.030           |
| → res (residual)              | 0.091          | 0.15   | 0.082              | <b>0.030</b>  | 0.033 | 0.042                     | 0.060              | 0.047           |
| → res-nplm (residual)         | 0.023          | 0.12   | 0.044              | <b>0.018</b>  | 0.074 | 0.037                     | 0.041              | 0.021           |
| NPLM-dist. (sup.)             | 0.040          | –      | 0.087              | <b>0.033</b>  | 0.084 | 0.049                     | 0.073              | 0.042           |
| → res (residual)              | 0.077          | 0.43   | 0.056              | <b>0.020</b>  | 0.029 | 0.046                     | 0.046              | 0.034           |
| → res-nplm (residual)         | 0.049          | –      | 0.089              | <b>0.046</b>  | 0.089 | 0.080                     | 0.049 <sup>†</sup> | 0.062           |
| NPLM-dist.+classwise SIGReg   | 0.025          | –      | 0.028 <sup>†</sup> | <b>0.020</b>  | 0.072 | 0.029 <sup>†</sup>        | 0.027 <sup>†</sup> | 0.036           |
| → res (residual)              | 0.084          | –      | 0.079              | 0.050         | 0.046 | <b>0.044</b>              | 0.045              | 0.050           |
| → res-nplm (residual)         | 0.079          | –      | 0.044              | 0.050         | 0.046 | <b>0.043</b>              | 0.057              | 0.050           |

**Table 6: Post-discovery  $f^*(2\sigma)$  on the density-ratio pool, derived cut.** The frozen-space SparKer  $f^*$  is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of  $d_{\text{seen}} - d_{\text{disc}}$  and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage*: CIFAR-10 holdout 4, density-ratio pool, derived cut; 12 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys,  $N_D=5000$ ; purity = round-1 pool purity at  $f=0.03$ ; <sup>†</sup> = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘–’ = declined at  $f=0.03$ ; an ‘(n/m)’ row is an arm still running.

| arm                           | frozen SparKer | purity | Eucl. | Eucl. (anch.) | Maha. | MMD   | SparKer | SparKer (anch.) |
|-------------------------------|----------------|--------|-------|---------------|-------|-------|---------|-----------------|
| SupCon                        | 0.021          | 0.00   | > 0.1 | > 0.1         | 0.043 | 0.047 | 0.052   | <b>0.035</b>    |
| SupCon+SIGReg ( $\lambda=5$ ) | 0.034          | 0.03   | 0.042 | <b>0.021</b>  | > 0.1 | 0.056 | 0.040   | 0.043           |
| NPLM-dist. (sup.)             | 0.029          | 0.08   | 0.082 | <b>0.022</b>  | 0.038 | 0.044 | 0.029   | 0.023           |
| NPLM-dist.+classwise SIGReg   | 0.029          | 0.17   | 0.029 | <b>0.018</b>  | 0.024 | 0.037 | 0.028   | 0.025           |

**Table 7: Post-discovery  $f^*(2\sigma)$  on the distance pool, 95th-pct. cut.** The frozen-space SparKer  $f^*$  is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of  $d_{\text{seen}} - d_{\text{disc}}$  and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 7, distance pool, 95th-pct. cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys,  $N_D=5000$ ; purity = round-1 pool purity at  $f=0.03$ ; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at  $f=0.03$ ; an ‘(n/m)’ row is an arm still running.

| arm                           | frozen SparKer | purity | Eucl. | Eucl. (anch.) | Maha. | MMD   | SparKer        | SparKer (anch.) |
|-------------------------------|----------------|--------|-------|---------------|-------|-------|----------------|-----------------|
| SupCon                        | 0.021          | 0.00   | 0.071 | > 0.1         | 0.036 | 0.056 | 0.040          | <b>0.033</b>    |
| SupCon+SIGReg ( $\lambda=5$ ) | 0.034          | –      | 0.079 | –             | 0.090 | 0.079 | <b>0.036</b> † | –               |
| NPLM-dist. (sup.)             | 0.029          | 0.05   | > 0.1 | <b>0.039</b>  | 0.044 | 0.042 | 0.042          | 0.041           |
| NPLM-dist.+classwise SIGReg   | 0.029          | 0.10   | 0.042 | 0.029         | 0.035 | 0.052 | 0.040          | <b>0.027</b>    |

**Table 8: Post-discovery  $f^*(2\sigma)$  on the distance pool, derived cut.** The frozen-space SparKer  $f^*$  is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of  $d_{\text{seen}} - d_{\text{disc}}$  and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 7, distance pool, derived cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys,  $N_D=5000$ ; purity = round-1 pool purity at  $f=0.03$ ; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at  $f=0.03$ ; an ‘(n/m)’ row is an arm still running.

| arm                           | frozen SparKer | purity | Eucl. | Eucl. (anch.) | Maha. | MMD   | SparKer        | SparKer (anch.) |
|-------------------------------|----------------|--------|-------|---------------|-------|-------|----------------|-----------------|
| SupCon                        | 0.021          | 0.68   | 0.022 | 0.020         | 0.021 | 0.052 | <b>0.018</b>   | 0.020           |
| SupCon+SIGReg ( $\lambda=5$ ) | 0.034          | –      | 0.079 | –             | 0.090 | 0.072 | <b>0.036</b> † | –               |
| NPLM-dist. (sup.)             | 0.029          | 0.18   | 0.044 | <b>0.021</b>  | 0.033 | 0.043 | 0.041          | 0.029           |
| NPLM-dist.+classwise SIGReg   | 0.029          | 0.67   | 0.043 | <b>0.011</b>  | 0.018 | 0.049 | 0.033          | 0.019           |

**Table 9: Post-discovery  $f^*(2\sigma)$  on the density-ratio pool, derived cut.** The frozen-space SparKer  $f^*$  is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of  $d_{\text{seen}} - d_{\text{disc}}$  and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 7, density-ratio pool, derived cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys,  $N_D=5000$ ; purity = round-1 pool purity at  $f=0.03$ ; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at  $f=0.03$ ; an ‘(n/m)’ row is an arm still running.

| arm                           | frozen SparKer | purity | Eucl. | Eucl. (anch.) | Maha. | MMD          | SparKer | SparKer (anch.) |
|-------------------------------|----------------|--------|-------|---------------|-------|--------------|---------|-----------------|
| SupCon                        | 0.018          | 0.00   | 0.039 | <b>0.016</b>  | 0.023 | 0.018        | 0.021   | 0.017           |
| SupCon+SIGReg ( $\lambda=5$ ) | 0.040          | 0.01   | 0.026 | 0.019         | > 0.1 | <b>0.015</b> | 0.026   | 0.023           |
| NPLM-dist. (sup.)             | 0.024          | 0.16   | 0.037 | 0.017         | 0.019 | 0.019        | 0.021   | <b>0.016</b>    |
| NPLM-dist.+classwise SIGReg   | 0.024          | 0.18   | 0.017 | <b>0.015</b>  | 0.022 | 0.022        | 0.017   | 0.015           |

**Table 10: Post-discovery  $f^*(2\sigma)$  on the distance pool, 95th-pct. cut.** The frozen-space SparKer  $f^*$  is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of  $d_{\text{seen}} - d_{\text{disc}}$  and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 8, distance pool, 95th-pct. cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys,  $N_D=5000$ ; purity = round-1 pool purity at  $f=0.03$ ; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at  $f=0.03$ ; an ‘(n/m)’ row is an arm still running.

| arm                           | frozen SparKer | purity | Eucl. | Eucl. (anch.) | Maha. | MMD          | SparKer | SparKer (anch.) |
|-------------------------------|----------------|--------|-------|---------------|-------|--------------|---------|-----------------|
| SupCon                        | 0.018          | 0.00   | 0.052 | > 0.1         | 0.038 | <b>0.017</b> | 0.031   | 0.027           |
| SupCon+SIGReg ( $\lambda=5$ ) | 0.040          | 0.01   | 0.024 | 0.076         | 0.062 | <b>0.021</b> | 0.029   | 0.029           |
| NPLM-dist. (sup.)             | 0.024          | 0.59   | 0.063 | <b>0.015</b>  | 0.025 | 0.023        | 0.023   | 0.017           |
| NPLM-dist.+classwise SIGReg   | 0.024          | 0.55   | 0.042 | <b>0.012</b>  | 0.039 | 0.036        | 0.035   | 0.021           |

**Table 11: Post-discovery  $f^*(2\sigma)$  on the distance pool, derived cut.** The frozen-space SparKer  $f^*$  is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of  $d_{\text{seen}} - d_{\text{disc}}$  and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 8, distance pool, derived cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys,  $N_D=5000$ ; purity = round-1 pool purity at  $f=0.03$ ; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at  $f=0.03$ ; an ‘(n/m)’ row is an arm still running.

| arm                           | frozen SparKer | purity | Eucl. | Eucl. (anch.) | Maha. | MMD   | SparKer | SparKer (anch.) |
|-------------------------------|----------------|--------|-------|---------------|-------|-------|---------|-----------------|
| SupCon                        | 0.018          | 0.45   | 0.014 | <b>0.011</b>  | 0.015 | 0.019 | 0.017   | 0.014           |
| SupCon+SIGReg ( $\lambda=5$ ) | 0.040          | 0.22   | 0.024 | <b>0.020</b>  | 0.024 | 0.025 | 0.032   | 0.035           |
| NPLM-dist. (sup.)             | 0.024          | 0.64   | 0.046 | <b>0.016</b>  | 0.041 | 0.024 | 0.018   | 0.017           |
| NPLM-dist.+classwise SIGReg   | 0.024          | 0.73   | 0.075 | <b>0.009</b>  | 0.037 | 0.044 | 0.040   | 0.021           |

**Table 12: Post-discovery  $f^*(2\sigma)$  on the density-ratio pool, derived cut.** The frozen-space SparKer  $f^*$  is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of  $d_{\text{seen}} - d_{\text{disc}}$  and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 8, density-ratio pool, derived cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys,  $N_D=5000$ ; purity = round-1 pool purity at  $f=0.03$ ; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at  $f=0.03$ ; an ‘(n/m)’ row is an arm still running.

| arm                           | frozen SparKer | purity | Eucl.        | Eucl. (anch.) | Maha.        | MMD   | SparKer | SparKer (anch.) |
|-------------------------------|----------------|--------|--------------|---------------|--------------|-------|---------|-----------------|
| SupCon                        | 0.018          | 0.02   | <b>0.008</b> | 0.010         | 0.017        | 0.018 | 0.010   | 0.009           |
| SupCon+SIGReg ( $\lambda=5$ ) | 0.032          | 0.01   | 0.025        | <b>0.016</b>  | 0.040        | 0.040 | 0.025   | 0.025           |
| NPLM-dist. (sup.)             | 0.024          | 0.14   | 0.017        | 0.014         | 0.015        | 0.018 | 0.014   | <b>0.009</b>    |
| NPLM-dist.+classwise SIGReg   | 0.028          | 0.17   | 0.016        | 0.022         | <b>0.016</b> | 0.029 | 0.017   | 0.017           |

**Table 13: Post-discovery  $f^*(2\sigma)$  on the distance pool, 95th-pct. cut.** The frozen-space SparKer  $f^*$  is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of  $d_{\text{seen}} - d_{\text{disc}}$  and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 9, distance pool, 95th-pct. cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys,  $N_D=5000$ ; purity = round-1 pool purity at  $f=0.03$ ; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at  $f=0.03$ ; an ‘(n/m)’ row is an arm still running.

| arm                           | frozen SparKer | purity | Eucl. | Eucl. (anch.) | Maha. | MMD            | SparKer | SparKer (anch.) |
|-------------------------------|----------------|--------|-------|---------------|-------|----------------|---------|-----------------|
| SupCon                        | 0.018          | 0.00   | 0.013 | <b>0.011</b>  | 0.025 | 0.020          | 0.014   | 0.018           |
| SupCon+SIGReg ( $\lambda=5$ ) | 0.032          | –      | 0.047 | > 0.1         | 0.048 | <b>0.026</b> † | 0.027†  | 0.050           |
| NPLM-dist. (sup.)             | 0.024          | 0.82   | 0.036 | <b>0.020</b>  | 0.024 | 0.029          | 0.021   | 0.027           |
| NPLM-dist.+classwise SIGReg   | 0.028          | 0.51   | 0.026 | <b>0.020</b>  | 0.028 | 0.036          | 0.026   | 0.020           |

**Table 14: Post-discovery  $f^*(2\sigma)$  on the distance pool, derived cut.** The frozen-space SparKer  $f^*$  is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of  $d_{\text{seen}} - d_{\text{disc}}$  and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 9, distance pool, derived cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys,  $N_D=5000$ ; purity = round-1 pool purity at  $f=0.03$ ; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at  $f=0.03$ ; an ‘(n/m)’ row is an arm still running.

| arm                           | frozen SparKer | purity | Eucl. | Eucl. (anch.) | Maha. | MMD            | SparKer      | SparKer (anch.) |
|-------------------------------|----------------|--------|-------|---------------|-------|----------------|--------------|-----------------|
| SupCon                        | 0.018          | 0.28   | 0.010 | 0.013         | 0.016 | 0.025          | <b>0.009</b> | 0.010           |
| SupCon+SIGReg ( $\lambda=5$ ) | 0.032          | –      | 0.047 | 0.050         | 0.045 | <b>0.026</b> † | 0.027†       | 0.050           |
| NPLM-dist. (sup.)             | 0.024          | 0.84   | 0.027 | <b>0.020</b>  | 0.023 | 0.029          | 0.023        | 0.020           |
| NPLM-dist.+classwise SIGReg   | 0.028          | 0.86   | 0.044 | <b>0.020</b>  | 0.025 | 0.038          | 0.025        | 0.020           |

**Table 15: Post-discovery  $f^*(2\sigma)$  on the density-ratio pool, derived cut.** The frozen-space SparKer  $f^*$  is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of  $d_{\text{seen}} - d_{\text{disc}}$  and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 9, density-ratio pool, derived cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys,  $N_D=5000$ ; purity = round-1 pool purity at  $f=0.03$ ; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at  $f=0.03$ ; an ‘(n/m)’ row is an arm still running.

| space                                      | probe | per-ev. | Maha. |      |      |      |      | MMD  |      |      |      |      | SparKer |      |      |      |      |
|--------------------------------------------|-------|---------|-------|------|------|------|------|------|------|------|------|------|---------|------|------|------|------|
|                                            |       |         | c2    | c6   | c5   | c4   | c3   | c2   | c6   | c5   | c4   | c3   | c2      | c6   | c5   | c4   | c3   |
| SupCon                                     | 0.892 | 0.05    | .029  | .052 | .078 | .037 | >.1  | .015 | .048 | .031 | .022 | .023 | .015    | .073 | .066 | .035 | .042 |
| SupCon+SIGReg ( $\lambda=5$ )              | 0.762 | 0.13    | .023  | >.1  | >.1  | .033 | >.1  | .015 | .029 | .029 | .040 | .037 | .014    | >.1  | >.1  | .028 | .043 |
| NPLM-dist. (sup.)                          | 0.787 | 0.13    | >.1   | >.1  | >.1  | .021 | >.1  | .019 | .063 | .037 | .026 | .024 | .029    | .032 | .087 | .022 | .031 |
| SimCLR                                     | 0.829 | 0.05    | .029  | .046 | >.1  | .036 | .070 | .043 | .039 | .070 | .045 | .047 | .078    | .081 | >.1  | .095 | >.1  |
| SimCLR+SIGReg                              | 0.836 | 0.06    | .048  | .065 | >.1  | .034 | >.1  | .035 | .029 | .039 | .080 | .062 | .046    | .095 | >.1  | .046 | .093 |
| NPLM-bilinear                              | 0.808 | 0.08    | .078  | .042 | >.1  | .026 | >.1  | .038 | .039 | .056 | .046 | .052 | .052    | .070 | .093 | .049 | .088 |
| GCD (transductive)                         | 0.899 | 0.30    | .009  | .013 | .009 | .031 | .017 | .019 | .025 | .029 | .024 | .019 | .009    | .017 | .021 | .031 | .025 |
| GCD+SIGReg (transd.)                       | 0.872 | 0.48    | .015  | .019 | .019 | .043 | .025 | .014 | .035 | .040 | .042 | .029 | .013    | .027 | .026 | .030 | .028 |
| SupCon $\rightarrow$ res (residual)        | 0.910 | 0.11    | .085  | .048 | >.1  | .024 | .025 | .021 | .040 | .045 | .043 | .039 | .036    | .063 | .072 | .044 | .040 |
| SupCon $\rightarrow$ res (concat)          | 0.945 | 0.12    | .084  | .041 | >.1  | .024 | .034 | .019 | .040 | .042 | .042 | .038 | .029    | .056 | .046 | .042 | .026 |
| SupCon $\rightarrow$ res-nplm (residual)   | 0.887 | 0.08    | >.1   | .027 | >.1  | .033 | >.1  | .016 | .041 | .073 | .045 | .032 | .029    | .072 | .089 | .040 | .078 |
| SupCon $\rightarrow$ res-nplm (concat)     | 0.931 | 0.09    | .070  | .034 | .070 | .034 | .052 | .015 | .042 | .031 | .026 | .024 | .023    | .063 | .073 | .031 | .041 |
| SupCon+SIGReg $\rightarrow$ res (residual) | 0.854 | 0.06    | >.1   | .043 | >.1  | .019 | .041 | .019 | .036 | .045 | .040 | .031 | .043    | .089 | .081 | .037 | .072 |
| SupCon+SIGReg $\rightarrow$ res (concat)   | 0.855 | 0.09    | .031  | .063 | >.1  | .021 | .042 | .017 | .026 | .037 | .036 | .031 | .019    | .092 | .085 | .028 | .041 |

**Table 16: Galaxy10 / lejepa: pre-discovery  $f^*(2\sigma)$ , every (space, draw) point.** ‘>.1’ = the test never reached  $2\sigma$  by  $f=0.1$  on that draw. *Coverage:* Galaxy10 / LeJEPA, 14 spaces  $\times$  5 single-holdout draws (columns named by the held-out class); exp-150 battery, 200 null / 50 signal toys,  $N_D=5000$ , fractions 0.006–0.1. GCD arms trained on the unlabelled corpus with the novel images present (transductive).

| test                                                                                                                             | d0                | d3                | d5                | d7                | d8                | median |
|----------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------|-------------------|-------------------|-------------------|--------|
| <i>SupCon</i> — engaged from d0: 0.01, d3: 0.02, d5: 0.01, d7: 0.01, d8: 0.02; median engaged purity 1.00                        |                   |                   |                   |                   |                   |        |
| Eucl.                                                                                                                            | .019              | .044              | .038              | .023              | .027              | .027   |
| Eucl. (anch.)                                                                                                                    | .010              | .020              | .018              | .014              | .020              | .018   |
| Maha.                                                                                                                            | .021              | .026              | .045              | .015              | .070              | .026   |
| MMD                                                                                                                              | .010              | .071              | .046              | .019              | .040              | .040   |
| SparKer                                                                                                                          | .010              | .047              | .044              | .028              | .026              | .028   |
| SparKer (anch.)                                                                                                                  | .010              | .036              | .042              | .025              | .025              | .025   |
| <i>SupCon+SIGReg</i> ( $\lambda=5$ ) — engaged from d0: 0.01, d3: 0.02, d5: 0.01, d7: 0.01, d8: 0.02; median engaged purity 1.00 |                   |                   |                   |                   |                   |        |
| Eucl.                                                                                                                            | .038              | .070              | .079              | .024              | .069              | .069   |
| Eucl. (anch.)                                                                                                                    | .010              | .020              | .016              | .014              | .020              | .016   |
| Maha.                                                                                                                            | .018              | .047              | .072              | .023              | .056              | .047   |
| MMD                                                                                                                              | .018              | .047              | .042              | .028              | .041              | .041   |
| SparKer                                                                                                                          | .017              | .062              | .084              | .019              | .043              | .043   |
| SparKer (anch.)                                                                                                                  | .010              | .052              | .069              | .018              | .028              | .028   |
| <i>NPLM-dist. (sup.)</i> — engaged from d0: never, d3: never, d5: 0.05, d7: 0.1, d8: 0.02; median engaged purity 0.59            |                   |                   |                   |                   |                   |        |
| Eucl.                                                                                                                            | >.1               | .070 <sup>†</sup> | >.1               | .021 <sup>†</sup> | .029 <sup>†</sup> | .070   |
| Eucl. (anch.)                                                                                                                    | —                 | —                 | —                 | —                 | .020              | .020   |
| Maha.                                                                                                                            | >.1               | >.1               | >.1               | .021 <sup>†</sup> | .045              | .150   |
| MMD                                                                                                                              | .016 <sup>†</sup> | .052 <sup>†</sup> | .037              | .027 <sup>†</sup> | .027 <sup>†</sup> | .027   |
| SparKer                                                                                                                          | .029 <sup>†</sup> | .028 <sup>†</sup> | .088 <sup>†</sup> | .023 <sup>†</sup> | .028 <sup>†</sup> | .028   |
| SparKer (anch.)                                                                                                                  | —                 | —                 | —                 | —                 | .020              | .020   |

**Table 17: Galaxy10 / lejepa: post-discovery  $f^*(2\sigma)$ , every (arm, test, draw) point.** <sup>†</sup> = the derived cut had declined at the crossing fraction (the space there is the frozen one); ‘>.1’ = never reached  $2\sigma$ . *Eucl. (anch.)* and *SparKer (anch.)* consume the discovered anchors. *Coverage:* Galaxy10 / lejepa, discovery mode: density-ratio pool, derived cut,  $n_{\min}=5$ , skip-on-refuse; head-only loop (2 rounds  $\times$  5 epochs) over the cached trunk banks; injection from the train bank (clamped at the holdout pool size); 200 null / 50 signal toys; 82 (test, draw) points.



| space                                      | probe | per-ev. | Maha. |      |      |      |      | MMD  |      |      |      |      | SparKer |      |      |      |      |
|--------------------------------------------|-------|---------|-------|------|------|------|------|------|------|------|------|------|---------|------|------|------|------|
|                                            |       |         | c2    | c6   | c5   | c4   | c3   | c2   | c6   | c5   | c4   | c3   | c2      | c6   | c5   | c4   | c3   |
| SupCon                                     | 0.906 | 0.10    | >.1   | .065 | >.1  | .042 | >.1  | .019 | .031 | .034 | .029 | .019 | .031    | .047 | .065 | .021 | .030 |
| SupCon+SIGReg ( $\lambda=5$ )              | 0.823 | 0.09    | >.1   | .065 | >.1  | .029 | >.1  | .025 | .018 | .033 | .037 | .033 | .036    | .046 | .076 | .027 | .043 |
| NPLM-dist. (sup.)                          | 0.738 | 0.13    | >.1   | .065 | >.1  | .019 | .052 | .024 | .065 | .043 | .028 | .029 | .032    | .040 | .094 | .017 | .024 |
| SimCLR                                     | 0.847 | 0.04    | .075  | .047 | >.1  | .031 | >.1  | .049 | .044 | .066 | .052 | .049 | .072    | .093 | .090 | .082 | >.1  |
| SimCLR+SIGReg                              | 0.835 | 0.05    | >.1   | .090 | >.1  | .038 | >.1  | .040 | .026 | .044 | .070 | .046 | .069    | .091 | >.1  | .042 | .092 |
| NPLM-bilinear                              | 0.672 | 0.06    | >.1   | .096 | >.1  | >.1  | >.1  | .044 | .033 | .034 | .059 | .034 | .069    | .063 | .047 | .097 | >.1  |
| GCD (transductive)                         | 0.886 | 0.40    | .015  | .009 | .012 | .041 | .018 | .024 | .021 | .026 | .029 | .028 | .010    | .015 | .017 | .027 | .019 |
| GCD+SIGReg (transd.)                       | 0.852 | 0.48    | .013  | .017 | .026 | .031 | .026 | .017 | .024 | .039 | .033 | .027 | .022    | .022 | .032 | .029 | .019 |
| SupCon $\rightarrow$ res (residual)        | 0.930 | 0.06    | >.1   | .037 | >.1  | .043 | .037 | .016 | .025 | .035 | .036 | .036 | .026    | .087 | .082 | .047 | .043 |
| SupCon $\rightarrow$ res (concat)          | 0.947 | 0.09    | >.1   | .040 | .084 | .031 | .042 | .019 | .026 | .034 | .026 | .025 | .025    | .045 | .063 | .028 | .029 |
| SupCon $\rightarrow$ res-nplm (residual)   | 0.890 | 0.11    | >.1   | .052 | >.1  | .024 | .073 | .019 | .036 | .033 | .047 | .027 | .031    | .065 | .081 | .022 | .029 |
| SupCon $\rightarrow$ res-nplm (concat)     | 0.923 | 0.12    | >.1   | .048 | >.1  | .033 | .075 | .018 | .035 | .033 | .029 | .024 | .029    | .049 | .071 | .023 | .029 |
| SupCon+SIGReg $\rightarrow$ res (residual) | 0.867 | 0.05    | >.1   | .056 | >.1  | .017 | .052 | .025 | .024 | .039 | .045 | .038 | .038    | .086 | .080 | .043 | .062 |
| SupCon+SIGReg $\rightarrow$ res (concat)   | 0.888 | 0.09    | >.1   | .042 | >.1  | .019 | .036 | .021 | .021 | .033 | .040 | .029 | .029    | .043 | .052 | .030 | .037 |

**Table 18: Galaxy10 / visreg: pre-discovery  $f^*(2\sigma)$ , every (space, draw) point.** ‘>.1’ = the test never reached  $2\sigma$  by  $f=0.1$  on that draw. *Coverage:* Galaxy10 / VISREG, 14 spaces  $\times$  5 single-holdout draws (columns named by the held-out class); exp-150 battery, 200 null / 50 signal toys,  $N_D=5000$ , fractions 0.006–0.1. GCD arms trained on the unlabelled corpus with the novel images present (transductive).

| test                                                                                                                             | d0                | d3   | d5                | d7                | d8                | median |
|----------------------------------------------------------------------------------------------------------------------------------|-------------------|------|-------------------|-------------------|-------------------|--------|
| <i>SupCon</i> — engaged from d0: 0.01, d3: 0.02, d5: 0.02, d7: 0.01, d8: 0.02; median engaged purity 1.00                        |                   |      |                   |                   |                   |        |
| Eucl.                                                                                                                            | .070              | .038 | .063              | .024              | .028              | .038   |
| Eucl. (anch.)                                                                                                                    | .015              | .020 | .021              | .019              | .020              | .020   |
| Maha.                                                                                                                            | >.1               | .024 | .056              | .021              | .042              | .042   |
| MMD                                                                                                                              | .017              | .037 | .040              | .029              | .027              | .029   |
| SparKer                                                                                                                          | .025              | .056 | .063              | .024              | .033              | .033   |
| SparKer (anch.)                                                                                                                  | .022              | .042 | .044              | .021              | .021              | .022   |
| <i>SupCon+SIGReg</i> ( $\lambda=5$ ) — engaged from d0: 0.01, d3: 0.01, d5: 0.01, d7: 0.01, d8: 0.02; median engaged purity 1.00 |                   |      |                   |                   |                   |        |
| Eucl.                                                                                                                            | .077              | .048 | >.1               | .016              | >.1               | .077   |
| Eucl. (anch.)                                                                                                                    | .017              | .022 | .016              | .013              | .020              | .017   |
| Maha.                                                                                                                            | >.1               | .026 | .097              | .023              | .059              | .059   |
| MMD                                                                                                                              | .019              | .036 | .039              | .041              | .034              | .036   |
| SparKer                                                                                                                          | .034              | .030 | .062              | .026              | .045              | .034   |
| SparKer (anch.)                                                                                                                  | .018              | .029 | .040              | .017              | .025              | .025   |
| <i>NPLM-dist. (sup.)</i> — engaged from d0: 0.02, d3: 0.05, d5: 0.1, d7: 0.03, d8: 0.1; median engaged purity 0.43               |                   |      |                   |                   |                   |        |
| Eucl.                                                                                                                            | .082 <sup>†</sup> | .041 | >.1               | .015 <sup>†</sup> | .026 <sup>†</sup> | .041   |
| Eucl. (anch.)                                                                                                                    | —                 | —    | —                 | .030              | —                 | .030   |
| Maha.                                                                                                                            | >.1               | .039 | >.1               | .016 <sup>†</sup> | .044 <sup>†</sup> | .044   |
| MMD                                                                                                                              | .018              | .044 | .044 <sup>†</sup> | .025              | .037 <sup>†</sup> | .037   |
| SparKer                                                                                                                          | .034 <sup>†</sup> | .032 | .087              | .017 <sup>†</sup> | .027 <sup>†</sup> | .032   |
| SparKer (anch.)                                                                                                                  | —                 | —    | —                 | .030              | —                 | .030   |

**Table 19: Galaxy10 / visreg: post-discovery  $f^*(2\sigma)$ , every (arm, test, draw) point.** <sup>†</sup> = the derived cut had declined at the crossing fraction (the space there is the frozen one); ‘>.1’ = never reached  $2\sigma$ . *Eucl. (anch.)* and *SparKer (anch.)* consume the discovered anchors. *Coverage:* Galaxy10 / visreg, discovery mode: density-ratio pool, derived cut,  $n_{\min}=5$ , skip-on-refuse; head-only loop (2 rounds  $\times$  5 epochs) over the cached trunk banks; injection from the train bank (clamped at the holdout pool size); 200 null / 50 signal toys; 82 (test, draw) points.

| space                                      | probe | per-ev. | Maha. |      |      |      |     | MMD  |      |      |      |      | SparKer |      |      |      |      |
|--------------------------------------------|-------|---------|-------|------|------|------|-----|------|------|------|------|------|---------|------|------|------|------|
|                                            |       |         | c2    | c6   | c5   | c4   | c3  | c2   | c6   | c5   | c4   | c3   | c2      | c6   | c5   | c4   | c3   |
| SupCon                                     | 0.879 | 0.06    | >.1   | >.1  | .063 | .076 | >.1 | .031 | .029 | .025 | .042 | .037 | .078    | >.1  | >.1  | .049 | .083 |
| SupCon+SIGReg ( $\lambda=5$ )              | 0.804 | 0.04    | >.1   | .072 | .069 | >.1  | >.1 | .021 | .025 | .031 | .047 | .029 | >.1     | >.1  | >.1  | .079 | >.1  |
| NPLM-dist. (sup.)                          | 0.775 | 0.10    | >.1   | >.1  | >.1  | .059 | >.1 | .037 | .023 | .029 | .038 | .037 | .054    | .086 | .070 | .038 | .048 |
| SimCLR                                     | 0.790 | 0.05    | >.1   | .066 | >.1  | >.1  | >.1 | .029 | .039 | .056 | .067 | .052 | .070    | .097 | >.1  | .066 | >.1  |
| SimCLR+SIGReg                              | 0.801 | 0.04    | >.1   | .069 | >.1  | .076 | >.1 | .027 | .029 | .041 | .088 | .059 | .069    | .084 | >.1  | .059 | .091 |
| NPLM-bilinear                              | 0.684 | 0.04    | >.1   | >.1  | .082 | >.1  | >.1 | .097 | .042 | .041 | .093 | >.1  | .081    | .089 | .079 | .083 | .089 |
| GCD (transductive)                         | 0.821 | 0.07    | >.1   | .052 | .037 | .076 | >.1 | .019 | .025 | .031 | .040 | .028 | .059    | .087 | .085 | .069 | .092 |
| GCD+SIGReg (transd.)                       | 0.831 | 0.12    | .084  | .066 | .056 | .073 | >.1 | .024 | .026 | .034 | .039 | .034 | .052    | >.1  | .096 | .066 | .098 |
| SupCon $\rightarrow$ res (residual)        | 0.891 | 0.05    | >.1   | .080 | .062 | .097 | >.1 | .023 | .029 | .047 | .056 | .033 | .090    | .098 | .097 | .081 | .097 |
| SupCon $\rightarrow$ res (concat)          | 0.903 | 0.06    | >.1   | >.1  | .063 | .074 | >.1 | .025 | .025 | .029 | .048 | .034 | .074    | >.1  | .098 | .073 | .092 |
| SupCon $\rightarrow$ res-nplm (residual)   | 0.862 | 0.07    | >.1   | >.1  | .056 | .087 | >.1 | .024 | .024 | .036 | .039 | .032 | .077    | >.1  | >.1  | .047 | .092 |
| SupCon $\rightarrow$ res-nplm (concat)     | 0.890 | 0.06    | >.1   | >.1  | .063 | .075 | >.1 | .029 | .026 | .027 | .042 | .033 | .048    | .093 | >.1  | .047 | .092 |
| SupCon+SIGReg $\rightarrow$ res (residual) | 0.852 | 0.06    | >.1   | .073 | .092 | >.1  | >.1 | .025 | .025 | .044 | .049 | .032 | .088    | >.1  | .096 | .082 | >.1  |
| SupCon+SIGReg $\rightarrow$ res (concat)   | 0.869 | 0.05    | >.1   | .079 | .072 | .096 | >.1 | .021 | .025 | .036 | .047 | .033 | .089    | >.1  | >.1  | .075 | >.1  |

**Table 20: Galaxy10 / dino: pre-discovery  $f^*(2\sigma)$ , every (space, draw) point.** ‘>.1’ = the test never reached  $2\sigma$  by  $f=0.1$  on that draw. *Coverage:* Galaxy10 / DINO, 14 spaces  $\times$  5 single-holdout draws (columns named by the held-out class); exp-150 battery, 200 null / 50 signal toys,  $N_D=5000$ , fractions 0.006–0.1. GCD arms trained on the unlabelled corpus with the novel images present (transductive).

| test                                                                                                                              | d0                | d3                | d5                | d7                | d8   | median |
|-----------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------|-------------------|-------------------|------|--------|
| <i>SupCon</i> — engaged from d0: 0.01, d3: 0.02, d5: 0.03, d7: 0.01, d8: 0.01; median engaged purity 1.00                         |                   |                   |                   |                   |      |        |
| Eucl.                                                                                                                             | >.1               | .090              | .022              | >.1               | >.1  | .150   |
| Eucl. (anch.)                                                                                                                     | .019              | .021              | .030              | .014              | .018 | .019   |
| Maha.                                                                                                                             | >.1               | >.1               | .066              | >.1               | >.1  | .150   |
| MMD                                                                                                                               | .029              | .039              | .036              | .059              | .044 | .039   |
| SparKer                                                                                                                           | .062              | .079              | >.1               | .048              | .079 | .079   |
| SparKer (anch.)                                                                                                                   | .044              | .044              | .089              | .035              | .042 | .044   |
| <i>SupCon+SIGReg</i> ( $\lambda=5$ ) — engaged from d0: 0.02, d3: 0.01, d5: 0.01, d7: never, d8: 0.01; median engaged purity 1.00 |                   |                   |                   |                   |      |        |
| Eucl.                                                                                                                             | >.1               | >.1               | .062              | >.1               | >.1  | .150   |
| Eucl. (anch.)                                                                                                                     | .020              | .019              | .022              | —                 | .031 | .021   |
| Maha.                                                                                                                             | >.1               | .075              | .078              | >.1               | >.1  | .150   |
| MMD                                                                                                                               | .027              | .029              | .039              | .046 <sup>†</sup> | .046 | .039   |
| SparKer                                                                                                                           | .048              | .077              | .084              | .075 <sup>†</sup> | >.1  | .077   |
| SparKer (anch.)                                                                                                                   | .044              | .046              | .047              | —                 | .052 | .047   |
| <i>NPLM-dist. (sup.)</i> — engaged from d0: never, d3: never, d5: never, d7: never, d8: 0.03; median engaged purity 0.28          |                   |                   |                   |                   |      |        |
| Eucl.                                                                                                                             | .071 <sup>†</sup> | >.1               | >.1               | .018 <sup>†</sup> | >.1  | .150   |
| Eucl. (anch.)                                                                                                                     | —                 | —                 | —                 | —                 | .030 | .030   |
| Maha.                                                                                                                             | >.1               | >.1               | >.1               | .072 <sup>†</sup> | >.1  | .150   |
| MMD                                                                                                                               | .035 <sup>†</sup> | .024 <sup>†</sup> | .029 <sup>†</sup> | .037 <sup>†</sup> | .052 | .035   |
| SparKer                                                                                                                           | .045 <sup>†</sup> | .078 <sup>†</sup> | .056 <sup>†</sup> | .045 <sup>†</sup> | .094 | .056   |
| SparKer (anch.)                                                                                                                   | —                 | —                 | —                 | —                 | .047 | .047   |

**Table 21: Galaxy10 / dino: post-discovery  $f^*(2\sigma)$ , every (arm, test, draw) point.** <sup>†</sup> = the derived cut had declined at the crossing fraction (the space there is the frozen one); ‘>.1’ = never reached  $2\sigma$ . *Eucl. (anch.)* and *SparKer (anch.)* consume the discovered anchors. *Coverage:* Galaxy10 / dino, discovery mode: density-ratio pool, derived cut,  $n_{\min}=5$ , skip-on-refuse; head-only loop (2 rounds  $\times$  5 epochs) over the cached trunk banks; injection from the train bank (clamped at the holdout pool size); 200 null / 50 signal toys; 80 (test, draw) points.